

Homework 9

Problem 1 (Infinite-dimensional projection). Let H be a Hilbert space. Show that if $v \in H$ and $U \leq H$ is a closed linear subspace of H , there exists a unique projection $\text{proj}_U(v) \in U$ which is the point in U that is the closest to v . Furthermore, show that

$$v - \text{proj}_U(v) \in U^\perp$$

and $U \oplus U^\perp = H$.

Hint: as a reminder, $U^\perp := \{v \in H : v \perp U\}$. We can not use the finite-dimensional projection formula from lecture since U is not necessarily finite-dimensional. You need to argue by Cauchy-completeness and the parallelogram law to show existence + uniqueness. Also recall what direct sum means.

Problem 2 (Representation theorem). Let H be a complex Hilbert space and $\phi : H \rightarrow \mathbb{C}$ be a bounded (continuous) linear functional. Assume that $\phi \neq 0$.

1. With $H_0 := \text{Ker}(\phi)$ (the kernel / nullspace of ϕ), show that $H_0^\perp$ is a one-dimensional subspace.

Hint: be careful that you need to argue $H_0^\perp \neq 0$.

2. Show that we can find $\alpha_\phi \in H$ such that

$$\phi(v) = \langle v, \alpha_\phi \rangle \quad \forall v \in V$$

Problem 3 (Sobolev embedding). Let $m, n \in \mathbb{N}$ and $f \in L^2(\mathbb{T}^n)$ (as a distribution).

1. Let $\alpha \in \mathbb{R}$. Show that

$$\|\langle k \rangle^\alpha\|_{L_k^2(\mathbb{Z}^n)} < \infty$$

if and only if $2\alpha < -n$.

2. Show that if $N > m + \frac{n}{2}$ and

$$\|f\|_{H^N} := \left\| \langle k \rangle^N \hat{f}(k) \right\|_{L^2(\mathbb{Z}^n)} < \infty$$

then $f \in C^m(\mathbb{T}^n)$ (there is a C^m function giving the same distribution as f) and

$$\|f\|_{C^m} \lesssim \|f\|_{H^N}$$

Hint: use Littlewood-Paley projection to “nicely” approximate f (in many norms) by smooth functions, and observe that a Cauchy sequence in H^N is also Cauchy in C^m .